

When Detection Metrics Cannot Discriminate: Acquisition Structure and Measurement-Shift Robustness in Industrial Anomaly Detection

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Abstract

Public predictive-maintenance benchmarks are increasingly used to justify deep sequence models over classical statistical monitors. We audit one such benchmark—a hydraulic-pump vibration and current dataset distributed with an official LSTM-Autoencoder reference analysis—and find that its headline comparison is not supported by the data. Three results drive this conclusion. First, the acquisition is *burst-sampled*: the nominally 10 Hz record consists of 599 bursts of at most 50 samples (4.9 s), so the reference pipeline’s 120-sample analysis span does not fit inside *any* burst, and its stated “10 s-ahead” horizon is unrealizable. Second, under a leakage-controlled protocol every model we test—including a one-line range rule—achieves $AP = 1.000$ and perfect recall, so detection metrics carry *zero* discriminative power on this benchmark; the published LSTM-AE score of $F_1 = 0.748$ —which we reproduce to within seed noise at 0.743—is below what a rule with no learned parameters attains ($F_1 = 0.973$), and the same network reaches 0.959 once placed inside a leakage-controlled protocol, so the published gap measures evaluation procedure rather than model capacity. Third, models that are indistinguishable on detection differ by more than $5\times$ in robustness to benign measurement shift: a pure DC offset applied to *normal* data raises the false-alarm rate by 89.7 points for a Mahalanobis monitor but only 17.0 points for PCA-MSPC. We argue that on benchmarks where normal and fault records are separable for trivial reasons—here a single fault event recorded on a different day—model selection must be driven by false-alarm behaviour under shift and by calibration quality, not by detection scores. We release a protocol (gap-aware windowing, burst-blocked normal-only cross-validation, conformal calibration, and shift-perturbation gates) that makes this selection explicit and reproducible in one command.

1 Introduction

Industrial predictive maintenance is an attractive application for deep sequence models: data are multivariate, temporal, and abundant, while labelled failures are rare. The standard recipe trains a reconstruction model on normal operation and thresholds reconstruction error (Malhotra et al., 2016; Audibert et al., 2020). Public datasets packaged with a reference implementation have become a common entry point for practitioners, and the reference numbers in those packages are frequently taken as the baseline to beat.

This paper is an audit of one such package. The dataset records vibration and current from the hydraulic pump of a fine-blanking press, and ships with a guidebook implementing an LSTM-Autoencoder that reports Accuracy = 97.51% and $F_1 = 0.748$ (Ministry of SMEs and Startups and KAIST, 2022). We set out to compare classical statistical monitors against this baseline under a common protocol. We did not obtain a comparison, because the benchmark does not support one. What we obtained instead is a case study in how acquisition artefacts and dataset construction can make a detection benchmark uninformative while still producing respectable-looking numbers.

Our contributions are as follows.

(1) The acquisition is not a continuous stream. We show (§3) that the record is burst-sampled: 599 bursts whose longest is 50 samples. The reference pipeline builds 20-sample windows with a 100-step label offset, a 120-sample span that fits inside no burst. Its windows therefore straddle gaps of arbitrary length, and the “10 s ahead” interpretation of the offset is an artefact of treating the row index as uniform time.

(2) Detection metrics cannot discriminate models here. Under a leakage-controlled protocol (§4), a one-line normal-range rule, a shrinkage Mahalanobis monitor, and PCA-MSPC all reach $AP = 1.000$, AUROC = 1.000 and recall 1.000 (§5). Normal and fault windows are linearly separable by a single feature. Reporting F_1 differences on this benchmark measures threshold placement, not detection ability.

(3) Robustness to benign measurement shift does discriminate, by a wide margin. Perturbing *normal* data with sensor gain, DC offset, polarity inversion and sampling jitter—changes that a maintenance system must not

report as equipment faults—separates the models cleanly (§6): worst-case false-alarm increase ranges from 17.0 to 89.7 percentage points. We make this the selection criterion.

(4) A reproducible protocol. We specify gap-aware windowing, burst-blocked normal-only block cross-validation, split-conformal calibration of the alarm threshold, and a sequence of selection gates that deliberately excludes detection score as a tie-breaker (§4). The full pipeline runs end-to-end in 25 s on a laptop CPU.

We state plainly what this benchmark cannot support. The fault class is a single continuous recording made five days after the normal recording, so perfect separability is confounded with date, operating condition and sensor state; and because the fault record begins after the equipment state has already changed, no lead-time claim is measurable. We treat these as properties of the benchmark to be reported, not obstacles to be modelled around.

2 Related Work

Evaluation pathologies in time-series anomaly detection. A growing literature argues that progress in this area is partly illusory. Kim et al. (2022) show that the widely used *point-adjustment* protocol inflates F_1 so severely that random scores appear state-of-the-art, and Wu and Keogh (2023) argue that several standard benchmarks are trivially solvable or mislabelled. Garg et al. (2022) reach similar conclusions in a systematic comparison. Our finding is adjacent but distinct: we do not use point adjustment, and the failure is not mislabelling. The benchmark is simply *too separable*, so every method saturates and the metric stops ordering methods. We therefore argue for replacing the ordering criterion rather than only repairing the metric.

Shortcut learning and confounded splits. That a model may key on a nuisance correlate rather than the phenomenon of interest is well documented in vision and medical imaging (Geirhos et al., 2020; Zech et al., 2018), where scanner or site identity can be predicted more easily than pathology; Recht et al. (2019) show how fragile benchmark conclusions can be to distribution details. The present dataset has the same structure in a particularly stark form: class is perfectly aligned with acquisition date.

Classical multivariate process monitoring. Hotelling’s T^2 with squared prediction error (SPE/ Q) is the standard tool for industrial monitoring (Jackson and Mudholkar, 1979; Qin, 2003), and provides per-variable contribution plots that support root-cause triage. We use it, shrinkage covariance estimation (Ledoit and Wolf, 2004), and Isolation Forest (Liu et al., 2008) as the comparison set against the reference LSTM-Autoencoder.

Table 1: Acquisition audit. “Span if continuous” is the duration the row count would occupy at the documented 10 Hz. The discrepancy is the total gap time.

	Normal	Fault
Rows	20,000	600
Observed span (s)	4615.8	165.6
Span if continuous (s)	2000.0	60.0
Median Δt (s)	0.100	0.100
Gaps > 0.5 s	598	20
Bursts	599	21
Burst length (median / max)	37 / 50	28 / 50
Bursts ≥ 120 samples	0	0
Missing values	0	0

Distribution-free calibration. We set alarm thresholds by split-conformal p -values (Vovk et al., 2005; Angelopoulos and Bates, 2023), which give a distribution-free statement about normal operation without ever consulting fault labels. Because overlapping windows violate exchangeability, we block by acquisition burst, in the spirit of purged cross-validation (López de Prado, 2018), and we report realized coverage rather than assuming it.

3 The Benchmark and Its Acquisition Structure

Dataset. The package (Ministry of SMEs and Startups and KAIST, 2022) contains two CSV files from a fine-blanking press hydraulic pump: a normal record of 20,000 rows (2022-07-12) and a fault record of 600 rows (2022-07-17). Each row carries upper and lower pump vibration (AI0, AI1), motor current (AI2), and a constant equipment-state flag—0 throughout the normal file and 1 throughout the fault file. Sampling is documented as 10 times per second.

The record is burst-sampled. Table 1 reports the audit. The normal file spans 4615.8 s of wall-clock time while containing only 2000 s worth of samples at 10 Hz; 598 inter-sample gaps exceed 0.5 s. Segmenting at those gaps yields 599 bursts with median 37 and maximum 50 samples—at most 4.9 s of contiguous signal. The fault file is identical in character: 21 bursts, maximum 50 samples. Figure 1 visualises the structure.

Consequence for the reference pipeline. The guidebook builds windows of 20 consecutive *rows* and pairs them with a label 100 rows later, describing this as detecting conditions 10 s ahead. Because no burst reaches 120 samples, that span always crosses at least one gap, and the realized time offset is not 10 s but an uncontrolled quantity determined by gap durations. Two consequences follow: the temporal-horizon claim is unrealizable, and a substantial fraction of “2-second”

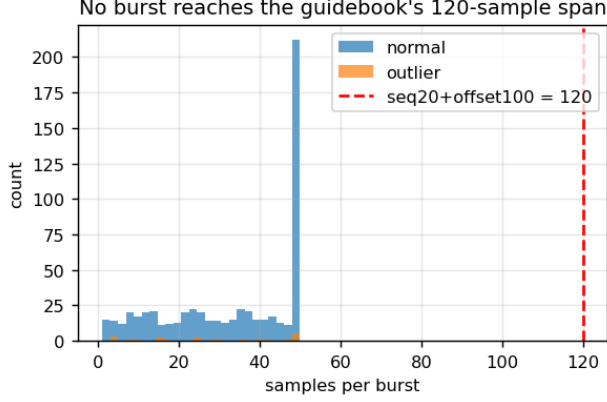


Figure 1: Burst length distribution. The reference pipeline’s analysis span (20-sample window + 100-step offset = 120 samples, dashed line) exceeds every burst in both files.

Table 2: Windows retained when windows are confined to a single burst, relative to naive index-contiguous windowing.

Length	Normal		Fault	
	Gap-aware	Retained	Gap-aware	Retained
5	17,652	88.3%	517	86.7%
10	14,871	74.4%	428	72.4%
20	9,978	49.9%	276	47.5%

windows are not contiguous in time at all. Enforcing that windows lie inside a single burst removes 25.6% of windows at length 10 and 50.1% at length 20 (Table 2); those removed windows are precisely the fictitious ones.

What the labels mean. The fault file is a single uninterrupted recording of an *already faulted* machine; the equipment-state flag is constant within each file. The 428 fault windows at length 10 are therefore overlapping fragments of *one* event, not 428 independent positives, and no pre-failure segment exists. We report window counts for transparency but treat the number of independent positive events as one throughout.

4 Protocol

We fix the following before examining any result; deviations made after seeing results are logged explicitly (§10).

Windowing. Windows are 10 samples with stride 1 and must lie entirely within one burst. Window length 5 and 20 are reported as sensitivity.

Features. From each window we compute four interpretable groups: **A** amplitude (per-channel standard deviation, peak-

to-peak, RMS), **S** shape (lag-1 and lag-2 autocorrelation, zero-crossing rate), **R** relation (AI0–AI1 correlation, absolute-value correlation, standard-deviation ratio), and **O** level (current mean and mean absolute value). A fifth group **Q** (sampling-interval variability, position within burst) is computed for diagnosis but excluded from model input so that acquisition artefacts cannot be mistaken for fault evidence. This yields a 23-dimensional model input.

Block cross-validation on normal data only. The normal record is partitioned into five contiguous blocks *by burst*, so no burst is split across blocks. Each fold uses three blocks for fitting, one for calibration, and one as normal hold-out, rotating so each block is hold-out exactly once. Random k -fold and window-level shuffling are prohibited: overlapping windows would place near-duplicates on both sides of the split.

Conformal alarm threshold. All scalars and thresholds are fit on the fitting and calibration blocks only. Given calibration scores $s_1, \dots, s_n$ from normal data, a new window receives

$$p_{\text{normal}}(x) = \frac{1 + |\{i : s_i \geq s(x)\}|}{n + 1}, \quad (1)$$

and we alarm when $p_{\text{normal}} \leq \alpha$ with $\alpha = 0.01$. We define risk = $1 - p_{\text{normal}}$ and emphasise that this is a rank statistic, not a posterior probability of failure; a prevalence sensitivity table is reported separately. Fault data are never used to choose thresholds, features or hyperparameters.

Models. **BL-0**, a range rule flagging any feature outside its training min–max; **M1**, shrinkage Mahalanobis distance (Ledoit and Wolf, 2004); **M2**, Isolation Forest (Liu et al., 2008); **M3**, PCA-MSPC combining T^2 and SPE (Jackson and Mudholkar, 1979); and **BL-1**, the reference LSTM-Autoencoder reproduced exactly ($64 \rightarrow 32 \rightarrow \text{repeat} \rightarrow 32 \rightarrow 64$, mean-squared reconstruction loss) in two variants defined in §8.

Metrics. We report window-level recall, precision, F_1 , MCC, average precision (AP) and AUROC; false-alarm window rate and merged alarm events per hour on each normal hold-out block, with a Poisson 95% upper bound so that a zero count is never reported as a rate of zero; realized conformal coverage; and false-alarm increase under measurement perturbation. We report the worst block, not only the mean.

5 Normal and Fault Are Perfectly Separable

Table 3 gives the five-fold result. BL-0, M1 and M3 all attain recall 1.000, AP = 1.000 and AUROC = 1.000; M2 is marginally lower. Every fault window is scored above

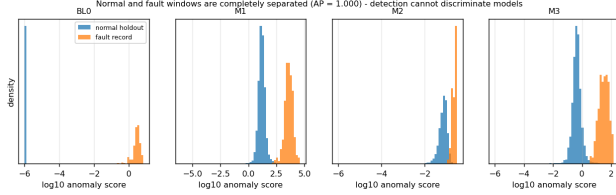


Figure 2: Anomaly-score distributions on fold 0. Normal hold-out and fault windows do not overlap for any monitor, including the parameter-free range rule BL-0. No threshold choice can distinguish these methods by detection.

Table 3: Five-fold block cross-validation. Detection metrics saturate; the models are ordered only by false alarms. FAR is per hour of acquisition on the worst hold-out block, with Poisson 95% upper bound. BL-1 is the reference LSTM-Autoencoder under our protocol (§8).

	Recall	AP	AUROC	FP rate (worst)	FAR ₉₅ (worst)
BL-0	1.000	1.000	1.000	0.0201	566.1
M1	1.000	1.000	1.000	0.0219	340.7
M2	0.932	0.983	0.997	0.0222	555.5
M3	1.000	0.999	1.000	0.0227	362.7
BL-1	1.000	1.000	1.000	0.0268	509.2

every normal hold-out window by a rule that merely checks whether features leave their training range.

Where the separability comes from. Ablating feature groups (Table 4) shows that shape features alone give $AP = 0.9986$ and amplitude features alone 0.9935 , while relation and current-level features alone give 0.69 and 0.51 . Removing any single group leaves AP above 0.98 : the signal is redundant across groups, which is the signature of a global difference between the two recordings rather than a localized fault mechanism.

Implication for the published baseline. Since a parameter-free range rule reaches $F_1 = 0.973$ under this protocol, the reference LSTM-Autoencoder’s reported $F_1 = 0.748$ cannot be read as evidence about sequence modelling. It is consistent with a protocol defect: the reference selects its threshold by intersecting precision and recall curves on a validation set that *contains 300 fault windows*, i.e. it uses fault labels to place the decision boundary of a method described as unsupervised, and it evaluates reconstruction error at only the final timestep of each window while training on the whole window. We revisit this in §8.

Table 4: Feature-group ablation (mean AP over folds). Any single group except R or O nearly saturates; the discriminative signal is redundant.

Feature set	M1	M3
All (A,S,R,O)	1.0000	0.9988
–A	0.9997	0.9984
–S	0.9987	0.9810
–R	1.0000	1.0000
–O	1.0000	0.9987
only A	0.9935	0.9772
only S	0.9986	0.9956
only R	0.6923	0.6922
only O	0.5052	0.5058

Table 5: Worst-case percentage-point increase in false-alarm rate when *normal* hold-out data undergo benign measurement perturbation. Lower is better. Recall drop on fault data is shown for reference.

	Gain	Offset	Polarity	Jitter	Recall drop
BL-0	61.3	30.0	0.5	16.3	0.0
M1	8.3	89.7	5.3	23.2	0.2
M2	22.3	5.0	1.9	6.6	2.6
M3	6.3	17.0	11.3	3.5	2.3

6 Robustness to Benign Measurement Shift

If detection metrics cannot order the models, something else must. A maintenance monitor’s central operational failure mode is raising equipment alarms for changes that are not equipment faults—a sensor re-seated with reversed polarity, a gain or offset drift, a timing jitter in the acquisition system. We test this directly: we perturb the *normal* hold-out data and measure how far the false-alarm rate rises relative to the unperturbed rate. The fault data are perturbed as well to check that detection survives.

Table 5 reports the worst case over perturbation magnitudes. The models that were indistinguishable in Table 3 differ by more than a factor of five. A DC offset of half a training standard deviation drives M1’s false-alarm rate from $\approx 2\%$ to 89.8% of normal windows; the range rule BL-0 collapses under a gain change. PCA-MSPC is the most stable, with a worst-case increase of 17.0 points. Meanwhile recall on fault data is essentially unaffected for BL-0, M1 and M3 (drop ≤ 2.4 points), precisely because recall is already saturated—confirming that the perturbation test is informative only on the normal side.

This result has a direct system-design consequence. Because a benign measurement change can produce a large anomaly score, a single-threshold alarm conflates two categories that demand different responses. We therefore emit a

Table 6: Realized conformal coverage $\Pr(p \leq \alpha)$ on normal hold-out, averaged over folds. BL-0’s score ties destroy p -value resolution.

α	0.01	0.05	0.10
BL-0	0.008	0.012	0.013
M1	0.010	0.047	0.098
M2	0.011	0.065	0.116
M3	0.012	0.052	0.113

three-level signal: *green* when $p_{\text{normal}} > \alpha$; *red* when both a full-feature monitor and a second monitor restricted to vibration-only features (excluding all current-derived features) agree, for three consecutive windows; and *yellow* otherwise—an alert whose recommended action is to verify sensor mounting, polarity, gain and acquisition timing before scheduling equipment inspection.

7 Calibration and Error Structure

Coverage. Equation 1 is exact under exchangeability; overlapping windows violate it. Table 6 reports realized coverage on normal hold-out blocks. M1, M2 and M3 track the nominal level within 0.017 at $\alpha = 0.10$. BL-0 does not: its realized coverage saturates at 0.013 regardless of α , because every window inside the training range receives an identical score of zero, so the p -value has no resolution below the tie mass. A model whose threshold cannot be read as a probability is unusable for alarm budgeting, irrespective of its detection score—a point invisible in Table 3.

Where false alarms occur. Decomposing hold-out errors by operating regime—defined by training-fold RMS terciles, never by fault data—shows that false alarms concentrate sharply in the low-load regime: 2.30% of low-load windows versus 0.20% of high-load windows, an $11.5\times$ ratio, rising to 4.69% on the worst block. Windows in the first second of a burst also alarm more often than later windows (1.53% vs. 0.94%). False negatives are zero in every stratum, a direct consequence of §5. The actionable reading is that this monitor’s errors are dominated by under-represented normal operating states, not by missed faults, and that additional *normal* data across load conditions would improve it more than additional fault data.

8 Reproducing the Reference LSTM-Autoencoder

We reproduce the reference architecture exactly and run it in two configurations. **G0** follows the guidebook: absolute-value preprocessing, index-contiguous 20-sample windows with the 100-step offset, min–max scaling fit on the first 15,000 normal rows, reconstruction error taken at the final timestep only, and a threshold chosen where precision and recall curves intersect on a validation set containing 300

fault windows. **G1** places the same architecture inside our protocol: raw signal, gap-aware windows, whole-window reconstruction error, and a conformal threshold fit on normal data only.

On the absolute-value step. The guidebook applies $|\cdot|$ to all three channels. This is worth isolating because it interacts with the preceding results. Window RMS is *exactly invariant* to this transform, since $|x|^2 = x^2$; so are mean absolute value and peak. What the transform does destroy is sign and phase structure—precisely the information a sequence model could exploit that a moment-based statistic cannot. Measured over the 14,871 normal windows, the transform leaves RMS, mean absolute value and peak *bitwise identical* (maximum change 0.0), while changing window standard deviation by up to 1.19×10^2 , signed mean by 2.78×10^2 , peak-to-peak by 3.04×10^2 and kurtosis by 4.88. The signed mean matters here: the fault record’s current channel carries a large DC offset relative to normal operation, and that offset is exactly what $|\cdot|$ confounds with an amplitude increase. The reference pipeline thus discards the only information that would justify the sequence model over a moment-based monitor, before training it. Empirically the discarded structure is not noise: the fault record is asymmetric in opposite directions on the two vibration channels (skewness +0.385 on AI0, −0.586 on AI1) whereas the normal record is near-symmetric on AI0 (+0.023).

Results. Table 7 reports both variants against the published values and against our selected monitor. Two things follow.

First, **G0 reproduces the published numbers**: averaged over three seeds we obtain $F_1 = 0.743 \pm 0.029$ against the published 0.748, with precision, recall and false-positive rate all within seed noise. The gap between the reference baseline and the parameter-free rule of §5 is therefore a property of the reference protocol, not an artefact of our reimplement.

Second, **the same network reaches $F_1 = 0.959$ once the protocol is corrected**. G1 attains recall 1.000 and AP = 1.000 on every fold, matching the feature-based monitors exactly. The improvement of +0.216 F_1 is attributable entirely to evaluation and preprocessing—gap-aware windows, raw signal, whole-window reconstruction error, and a threshold fit without fault labels—since the architecture, optimiser and training budget are unchanged. Even so, G1 does not overtake the one-line range rule, and it incurs the highest worst-block false-alarm rate of any model we test (0.0268 against 0.0201 for that rule). Note that the two halves of Table 7 are not directly comparable: the published and G0 rows come from the guidebook’s single held-out split, whereas the lower rows are five-fold means. We give both the mean and the worst block for the latter so that the comparison is not read off a favourable fold.

Table 7: Reference LSTM-Autoencoder. G0 reproduces the guidebook protocol; G1 places the same network in ours. The published row and G0 use the guidebook’s single train/test split, so no per-block worst case exists for them; rows below the rule are five-fold block cross-validation, reported as mean and worst block.

Published (Ministry of SMEs and Startups and KAIST, 2022)	0.664
G0 (reproduction, 3 seeds)	0.667
G1 (our protocol, 5 folds)	0.922
BL-0 (one-line rule)	0.949
M3 (selected)	0.928

We also note that under G0 early stopping never fired: all three seeds consumed the full 800-epoch budget despite a patience of 40 on training loss, whereas G1 stopped between 499 and 679 epochs. The reference configuration’s patience of 120 was therefore inactive.

9 Model Selection

We apply gates in a fixed order and deliberately exclude detection score as a tie-breaker: (i) numerical validity; (ii) robustness, using the normal-side false-alarm increase of §6 with a 20-point limit; (iii) worst-block false-alarm rate no worse than BL-0 within tolerance; (iv) at least one fault burst detected; (v) conformal coverage error ≤ 0.05 at $\alpha = 0.10$. Among survivors we prefer the lowest worst-block Poisson upper bound on alarms per hour.

Only **M3 (PCA-MSPC)** passes. BL-0 fails (ii) and (v); M1 fails (ii) on offset with +89.7 points; M2 fails (ii) on gain; BL-1 (G1) fails (iii) with the highest worst-block false-alarm rate of the set (0.0268 against a limit of 0.0251). The selected system uses M3 as the first stage and a vibration-only Mahalanobis monitor as the confirmation stage, which also supplies per-feature contributions for root-cause text. Top-contributing features are stable across folds (Jaccard overlap 0.87 for M3, 0.73 for M1).

We flag one asymmetry. Gates (ii) and (v) were *not* evaluated for BL-1: the perturbation study requires retraining for every fold $\times$ perturbation combination, which at the measured cost of this network (a single five-fold run takes 3.6 h on CPU) exceeds our compute budget by more than an order of magnitude. BL-1’s exclusion therefore rests on a single measured gate, and we report it as such rather than claiming it was dominated across all criteria.

We stress that this ranking would have been invisible—and the choice would have defaulted to the highest- F_1 model, BL-0—had we used detection metrics.

10 Amendments Made After Seeing Results

Three protocol changes were made after first results and are recorded here for completeness. (1) The robustness gate was originally defined on fault-side recall drop; with recall saturated at 1.000 this quantity was identically zero for three of four models and carried no information, so the normal-side false-alarm increase was added. (2) The false-alarm gate “no worse than BL-0” was given a tolerance (relative 10% of absolute 0.005); taken literally it eliminated every model by hundredths of a percentage point and reinstated the floor baseline as the winner. (3) The coverage gate (§7) did not exist in the original specification and was added once BL-0 was inspected during exploratory work preceding this protocol, so these results should not be read as a fully held-out evaluation.

11 Limitations

The fault class is one continuous recording; the 428 fault windows are overlapping fragments of a single event, so no confidence interval computed over windows reflects uncertainty about *faults*. Normal and fault were recorded on different days (2022-07-12 and 2022-07-17), so the perfect separability of §5 is confounded with date, product, tooling and sensor state; our perturbation study shows such nuisance factors are individually sufficient to move these models a great deal, but it cannot decompose the observed gap. No pre-failure segment exists, so lead time is not measurable and we make no early-warning claim. Finally, at 10 Hz the Nyquist limit is 5 Hz, far below gear-mesh and bearing fault frequencies (Randall and Antoni, 2011); we therefore avoid any spectral-diagnostic interpretation, and note that this also removes much of the motivation for a high-capacity temporal model.

12 Conclusion

A benchmark on which every method scores perfectly is not an easy benchmark; it is an uninformative one. On the dataset studied here, detection metrics order methods no better than chance, while robustness to benign measurement shift separates them by a factor of five and calibration quality eliminates the apparent winner outright. We recommend that packaged reference analyses for industrial datasets report acquisition structure, a parameter-free baseline, and a measurement-shift stress test alongside any headline score. In this instance all three would have changed the conclusion.

Reproducibility Statement

All results are produced by a single command on CPU (`python run_all.py --mode quick, 25s`) with a fixed seed; the deep baseline is enabled by `--mode full`. Input file SHA-256 digests, configuration and environment

are written to a run manifest. Three invariant tests are enforced: no window may cross an acquisition gap, no burst may be split across cross-validation blocks, and no fitted object may observe hold-out data. Every number in this paper is emitted to a CSV file named in the accompanying code.

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